Quantum Gravitation

Lecture 10

Yuran Seo (Physics Department)

KAIST

The 3.5 K Thermal Medium

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$\rho_T = Z^{-1} e^{-H/k_B T_m}$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$n_B(\omega) = 1/(e^{\hbar\omega/k_B T_m} - 1)$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$T_m = 3.5K$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

The 3.5 K Thermal Medium

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Reading: Seo, ch. 10; Thermal Medium Standard TMS-35; Q-Medium thermal lab guide